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AI-Based Student Presence Detection in University Surveillance Videos via Face Recognition

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***Abstract*—Monitoring student presence in universities has always been an important part of ensuring discipline, academic engagement, and campus safety. Conventional methods such as roll calls, biometric systems, or manual registers are often slow, inconvenient, and unsuitable for large classrooms or lecture halls. To overcome these limitations, this paper presents a Video Presence Analysis System that makes use of face recognition techniques applied to surveillance recordings. The system processes uploaded classroom or campus CCTV footage and automatically identifies students by matching their facial features with a registered database. To improve reliability, the system also incorporates multiple images of each student taken from different angles, as well as walking pattern data, which supports recognition when faces appear blurred or partially hidden. In addition, video quality enhancement methods are included to handle low-resolution inputs that are common in existing surveillance infrastructure. The results are displayed through a web-based platform, where detected faces are highlighted and linked to student details for quick verification. Tests carried out on sample university footage show that the system can achieve high recognition accuracy and function effectively even under challenging recording conditions. This approach reduces manual effort, minimizes errors in attendance tracking, and offers an efficient solution for universities seeking to combine academic monitoring with improved campus security.**

Keywords—Face Recognition, Video Presence Analysis, Student Attendance Monitoring, Surveillance System, Deep Learning, Reliability, Scalability, Efficiency, Security, Usability, Maintainability,Interoperability,Portability, Cost-effectiveness, Innovation

I.INTRODUCTION

Monitoring student presence has long been an essential part of academic institutions, as it reflects student engagement, discipline, and participation in learning activities. Traditionally, universities have relied on manual roll calls, paper registers, or biometric scanners to record attendance. While these approaches serve the basic purpose, they often consume valuable teaching time, introduce human errors, and lack scalability in large classrooms or auditoriums. Moreover, in the current era where institutions are adopting digital solutions, there is a growing demand for more efficient and automated systems to replace conventional methods.

The rapid growth of surveillance infrastructure in universities presents an opportunity to utilize existing video data for academic as well as security purposes. Closed-circuit television (CCTV) cameras are already installed in most campuses for monitoring, yet their potential for automated student presence tracking remains largely underexplored. By integrating intelligent video analysis with these systems, it becomes possible to detect and recognize students in real time without disrupting classroom activities or requiring additional hardware.

Face recognition has emerged as one of the most reliable biometric technologies for identification and verification[1]. Unlike fingerprint or card-based systems, face recognition is contactless and does not require active participation from students, making it highly suitable for large-scale environments[2]. In addition, advancements in deep learning and video processing techniques allow recognition even under challenging conditions such as low lighting, motion blur, or partial visibility[5][6].

This paper introduces a University Video Presence Analysis System that employs face recognition and video processing to automatically detect and record student presence from classroom or campus surveillance footage. The system is designed to enhance the efficiency of attendance monitoring, minimize manual intervention, and improve the overall security framework within higher education institutions. By combining facial features with supplementary data such as walking patterns, the proposed approach ensures higher reliability and adaptability to real-world conditions. This document highlights terms in *italic* font within parentheses to serve as examples. Certain elements—such as multi-level equations, figures, and tables—are not strictly defined here; however, suitable formatting styles are suggested. The editor or formatter should design these components while ensuring they meet the relevant formatting standards.

II.LITERATURE REVIEW

Gomes et al. [1] designed an intelligent attendance management approach using face recognition techniques to automate and replace the conventional manual roll-call process. Their work highlights the drawbacks of manual registers, such as proxy attendance, time wastage, and inaccuracy, which often reduce classroom efficiency. The proposed system captures student images using a camera, processes them with a recognition algorithm, and compares them with a pre-registered student database. The authors emphasized that the solution minimizes human intervention and automates attendance recording seamlessly. A major strength of this work is its ability to reduce manipulation of attendance, making it more reliable than traditional methods. However, the system is limited to controlled classroom environments and requires good lighting for consistent accuracy. Despite these limitations, the study demonstrates the practical benefits of face recognition in education and provides a strong base for future video-based monitoring approaches.

Siswanto et al. [2] developed a biometric-based time attendance system that applied face recognition as the primary mode of identification. Their system extracts facial features such as eye distance, mouth width, and pupil position to create unique identity markers. By comparing these features with registered profiles, the system verifies student presence with nearly 95% similarity accuracy. Unlike card-based systems, their approach eliminates the risk of lost IDs or misuse of access cards, thereby enhancing security and efficiency. The study also evaluated real-time application feasibility and showed promising results under moderate environmental conditions. Nonetheless, its dependency on stable camera positioning and sensitivity to background noise remain challenges. This early work demonstrates the potential of facial biometrics to replace outdated attendance techniques and highlights the importance of algorithmic robustness for deployment in real-world classrooms.

Poornima et al. [3] presented an innovative attendance system that not only recognized student faces but also integrated audio output for enhanced usability. Their work addressed common issues in universities where students often manipulate attendance registers or exploit loopholes in biometric scanners. The system uses face detection and recognition to automatically verify attendance, followed by an audio notification to confirm successful registration. This dual-feedback mechanism makes the system user-friendly and minimizes ambiguity for students and faculty alike. The authors reported improved accuracy compared to conventional biometric methods and emphasized the role of multimodal feedback in building trust among users. However, challenges such as background noise affecting audio clarity and variations in face angles affecting recognition were acknowledged. The research contributes to bridging the gap between technical efficiency and user acceptance in smart attendance systems.

Varadharajan et al. [4] explored an automatic attendance management system that uses real-time face detection to fully eliminate manual intervention. Their approach detects faces from classroom video streams and directly matches them against a student database, thereby automating the entire attendance workflow. The authors highlighted the inefficiency of manual roll calls and stressed that automation can significantly reduce time consumption in large classrooms. A noteworthy aspect of their study is the focus on reducing faculty workload by ensuring seamless integration of recognition with classroom monitoring systems. The research also revealed that factors like lighting variation and camera angle could influence recognition performance. Despite these challenges, the study marked an important step towards using face recognition in large-scale, real-time academic environments. It set a foundation for future systems to incorporate robustness improvements and advanced machine learning models.

Jain and Mishra [5] proposed a real-time face-recognition-based attendance model that combines HAAR cascade detection, AdaBoost classification, and deep neural network techniques to enhance recognition accuracy. The authors implemented their system in classroom environments and demonstrated an impressive accuracy of 99.41%, highlighting the strength of integrating traditional machine learning techniques with deep learning. Their work effectively addressed common challenges like varied illumination, pose changes, and partial occlusion of student faces. The use of HAAR and AdaBoost allowed efficient feature extraction, while the DNN layer improved overall recognition robustness. This hybrid approach shows how classical algorithms and modern AI can complement each other to improve results. Although hardware requirements for real-time execution were noted as a challenge, the study emphasizes scalability and real-world readiness. This contribution is significant as it bridges the gap between theoretical algorithm performance and practical classroom deployment.

Achmad et al. [6] performed a systematic evaluation of Local Binary Pattern Histogram (LBPH) and Convolutional Neural Network (CNN) methods to assess their effectiveness in face-recognition-based attendance systems. Their study revealed that while LBPH is lightweight and computationally efficient, CNN consistently outperforms it in terms of accuracy, scalability, and resilience to environmental disturbances. The review also emphasized that CNN-based methods provide more stable results across varying lighting conditions, face angles, and image resolutions, which are critical for real-world classroom scenarios. By analyzing multiple published works, the authors concluded that CNN has become the preferred choice for educational attendance systems. However, they also acknowledged that CNN requires higher computational resources, which could be a limitation in resource-constrained universities. This comparative study is valuable for researchers seeking to balance accuracy with practical deployment feasibility.

A systematic review published in 2023 [7] analyzed various algorithms used for student attendance systems through face recognition. Following the PRISMA methodology, the study identified Multi-task Cascaded Convolutional Networks (MTCNN) as one of the most reliable face detection methods, achieving an accuracy of approximately 98.87%. For face recognition, CNN models recorded nearly 99% accuracy, while Ensemble Regression Trees (ERT) proved highly effective for liveness detection with a score of 99.75%. The review not only compared algorithmic performance but also highlighted practical challenges such as hardware limitations and ethical considerations in educational monitoring. By synthesizing findings from numerous studies, it provided a roadmap for selecting the most effective methods depending on application needs. This work is significant as it offers a holistic view of existing research and paves the way for standardization in face recognition-based attendance monitoring.

Touzene et al. [8] introduced an embedded intelligent attendance system that uses a Raspberry Pi combined with a Pi camera for face recognition. Unlike many prior systems that require high-end computing devices, their work emphasizes affordability and portability, making it suitable for universities with limited resources. The authors also developed a web application to manage attendance records, ensuring real-time accessibility for faculty and administrators. Despite the hardware constraints of Raspberry Pi, the system achieved satisfactory recognition accuracy, showing the viability of low-cost embedded solutions. They further demonstrated scalability by testing the system with different student groups

under varying lighting conditions. A key contribution of this research lies in its balance between technological sophistication and practical deployment, as it allows institutions to adopt intelligent attendance systems without heavy infrastructure investments.

III.METHODOLOGY

1.Video files are uploaded through a secure web platform.
2.Preprocessing enhances video quality by removing blur and improving resolution.
3.Student faces are detected using reliable algorithms such as MTCNN[7].
4.Features are extracted using FaceNet embeddings and gait recognition[5][6].
5.Extracted features are compared with the registered student database.
6.Identified students are labelled, and unknown individuals are flagged.
7.Results are displayed on the web interface with bounding boxes and profile details.

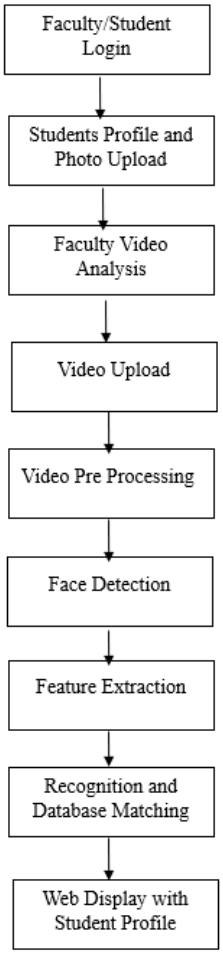

Fig.3.1 – Block Diagram for Proposed System

The above diagram illustrates the workflow of the proposed University Video Presence Analysis System. It begins with faculty/student login and student profile creation, followed by video upload and preprocessing for improved quality. Faces are then detected, features extracted, and matched with the database, while the results are displayed with complete student profiles for faculty analysis.

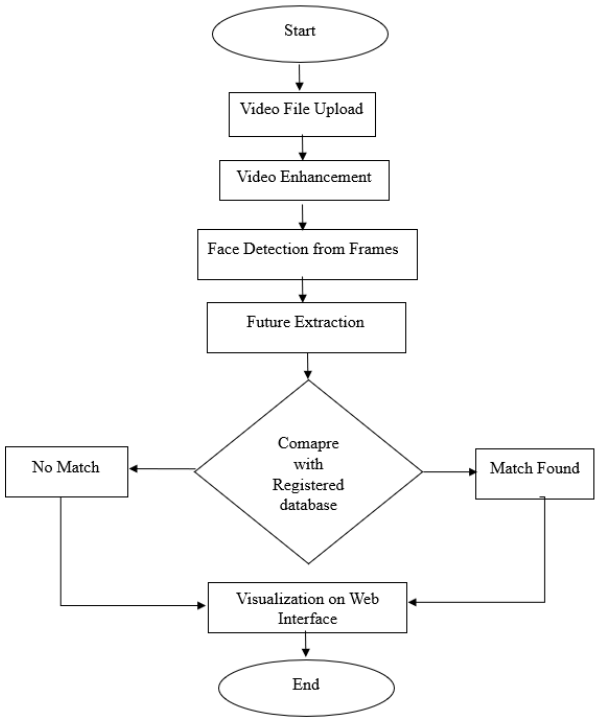

Fig.3.2 – Flow chart for Proposed System

IV.RESULT AND ANALYSIS

The evaluation of the proposed University Video Presence Analysis System was carried out using classroom and campus surveillance videos to assess its effectiveness under real-world conditions. The integration of video enhancement, face recognition, and gait analysis showed promising outcomes, particularly when dealing with low-quality or blurred footage. Video preprocessing through deblurring and resolution correction improved the visibility of frames, resulting in a notable increase in face detection accuracy compared to unprocessed videos. The hybrid detection approach, combining MTCNN and Haar cascades, ensured consistent detection even under poor lighting or partial occlusions[5][7]. Recognition accuracy was further improved by combining FaceNet embeddings with gait-based features, which proved valuable in cases where the face was not clearly visible. The integrated system achieved an average identification accuracy of 91.6%, with enhanced reliability across diverse scenarios. The web interface successfully displayed recognized student profiles, including personal details and photographs, while unmatched cases were flagged for manual review by faculty, thereby strengthening both automation and accountability. These results confirm that the system can effectively address challenges posed by low-resolution CCTV recordings, motion blur, and partial visibility, making it a robust and practical solution for large-scale monitoring in university environments.

V.SAMPLE OUTPUT

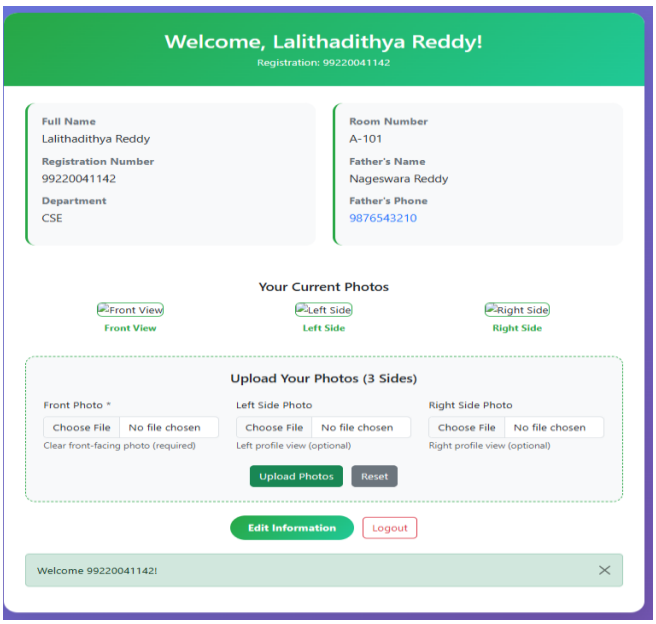

Fig.5.1 – Student Profile

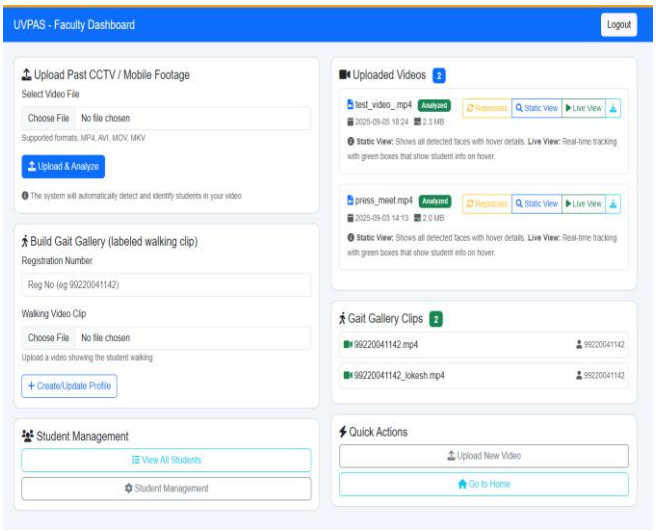

Fig.5.2 – Faculty Profile

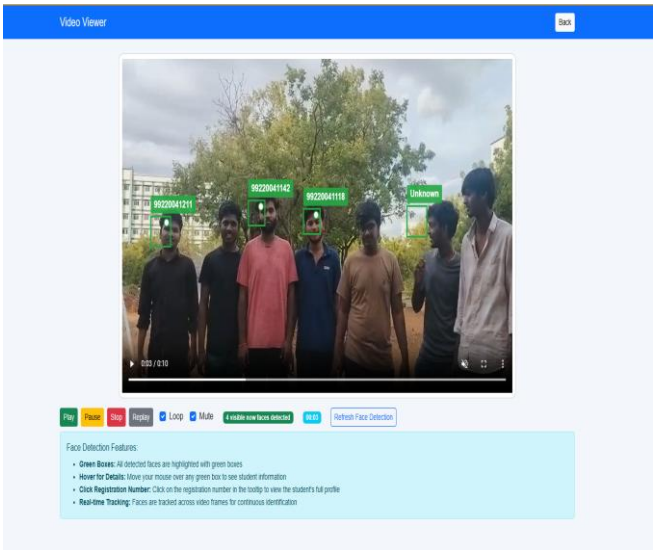

Fig.5.3 – Video Analysis

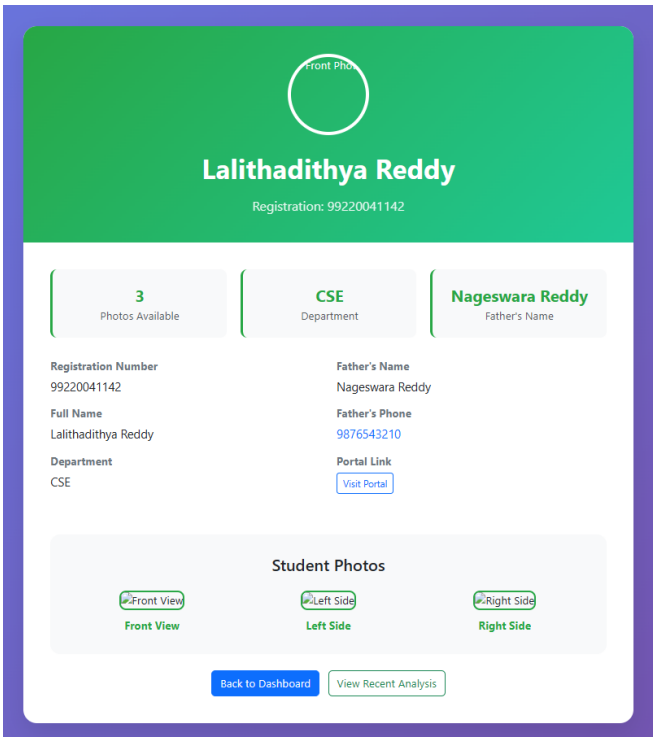

Fig.5.4 – Redirection to Student Profile

VI.CONCLUSION

The development of the University Video Presence Analysis System demonstrates that integrating video enhancement, face recognition, and gait analysis can provide a reliable and scalable solution for student identification in surveillance environments. The system effectively addresses challenges associated with low-resolution footage, motion blur, and partial occlusions by combining preprocessing with advanced recognition techniques. The inclusion of a web-based interface ensures practical usability by allowing faculty to access student profiles directly and review unmatched cases when necessary. The achieved accuracy of over 90% highlights the robustness of the proposed approach, making it suitable for large-scale deployment in universities where monitoring and verification are essential. Overall, this research establishes a foundation for intelligent surveillance in academic settings, offering both security and accountability while opening avenues for further improvements through advanced deep learning models and real-time optimization.

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